

Cal 2 - Exam Questions - Solution with Steps and SubSteps

Question 1

Let $f(x) = x^5 + x^3 - 4$. Find the value of $\frac{d}{dx}f^{-1}(x)$ at the point $x = -2$.

- a. $-\frac{1}{44}$
- b. $\frac{1}{92}$
- c. $\frac{1}{8}$
- d. I do not know

Solution: c) $\frac{1}{8}$

Step 1: Find $f'(x)$ (derivative of $f(x)$).

Step 1.1:

Apply the power rule to $f'(x)$ **Power Rule:** if $g(x) = x^n$, then $g'(x) = nx^{n-1}$.

Step 1.2:

Differentiate x^5 $\frac{d}{dx}(x^5) = 5x^4$

Step 1.3:

Differentiate x^3 $\frac{d}{dx}(x^3) = 3x^2$

Step 1.4:

Differentiate the constant -4 $\frac{d}{dx}(-4) = 0$

Step 1.5:

Combine the results $f'(x) = 5x^4 + 3x^2$

Step 2: Find $f^{-1}(-2)$ by using **Inverse Function : if $f(a) = b$ then $f^{-1}(b) = a$**

Step 2.1:

Set up the equation $f(x) = -2$. $x^5 + x^3 - 4 = -2$

Step 2.2:

Solve for x:

$$\begin{array}{ll} x^5 + x^3 - 4 = -2 & \text{Original equation.} \\ x^5 + x^3 - 4 + 4 = -2 + 4 & \text{Add 4 to each side.} \\ x^5 + x^3 = 2 & \text{Simplify both sides.} \end{array}$$

Step 2.3:

Test simple value 1 to find solution

$$\begin{array}{ll} f(x) = x^5 + x^3 - 4 & \\ f(1) = 1^5 + 1^3 - 4 & \text{Try } x = 1 \text{ in } f(x). \\ f(1) = 1 + 1 - 4 & \text{Simplify.} \\ f(1) = -2 & \end{array}$$

Step 2.4:

Conclude

Since $f(1) = -2$, $f^{-1}(-2) = 1$ due to Inverse Function

Step 3: Evaluate f' at $f^{-1}(-2)$

Step 3.1:

Substitute $x = 1$ into $f'(x)$

$$\begin{array}{ll} f'(x) = 5x^4 + 3x^2 & \\ f'(1) = 5(1)^4 + 3(1)^2 & \text{Substitute } x \text{ with 1 in } f'(x) \\ f'(1) = 5(1) + 3(1) & \\ f'(1) = 5 + 3 & \\ f'(1) = 8 & \end{array}$$

Step 4: Apply the inverse function derivative formula

Step 4.1:

Substitute into the formula

$$\frac{d}{dx}f^{-1}(x) = \frac{1}{f'(f^{-1}(x))}$$

Original Problem Equation.

$$\frac{d}{dx}f^{-1}(x) = \frac{1}{f'(f^{-1}(-2))}$$

Substitute x with -2

$$\frac{d}{dx}f^{-1}(x) = \frac{1}{f'(1)}$$

$$f'(1) = 8$$

$$\frac{d}{dx}f^{-1}(x) = \frac{1}{8}$$

Step 5: Final Answer: the correct option is C. $\frac{1}{8}$

Question 2

Evaluate $\int \frac{15x^4}{x^5+15} dx$.

- a. $\frac{3x^5}{\frac{1}{6}x^6+15x} + C$
- b. $15 \ln|x| + \frac{1}{5}x^4 + C$
- c. $3 \ln|x^5 + 15| + C$
- d. I do not know

Solution: c) $3 \ln|x^5 + 15| + C$

Step 1: Use U Substitution, Let $u = x^5 + 15$

Step 1.1:

Differentiate both sides of the equation $\frac{du}{dx} = \frac{d}{dx}(x^5 + 15)$

Step 1.2:

Differentiate x^5 $\frac{d}{dx}(x^5) = 5x^4$

Step 1.3:

Differentiate the constant 15 $\frac{d}{dx}(15) = 0$

Step 1.4:

Combine the results $\frac{d}{dx}(x^5 + 15) = 5x^4$

Step 1.5:

Multiply both sides by dx $du = 5x^4 dx$

Step 2: Rewrite $\int \frac{15x^4}{x^5+15} dx$ in terms of u.

Step 2.1:

Rewrite numerator $15x^4 dx = 3(5x^4 dx) = 3du$ since $du = 5x^4 dx$

Step 2.2:

Rewrite denominator $x^5 + 15 = u$

Step 2.3:

Combine the fraction $\int \frac{15x^4}{x^5+15} dx = \int \frac{3}{u} du$

Step 3: Evaluate $\int \frac{3}{u} du$

Step 3.1:

Factor out the constant

$$\int \frac{3}{u} du = 3 \int \frac{1}{u} du$$

Step 3.2:

Evaluate the integral

$$\int \frac{1}{u} du = \ln |u| + C$$

Step 3.3:

Combine the solution

$$\int \frac{3}{u} du = 3 \int \frac{1}{u} du = 3(\ln |u|) + C = 3 \ln |u| + C$$

Step 4: Substitute back to x

Step 4.1:

Replace u with $x^5 + 15$

$$3 \ln |u| + C = 3 \ln |x^5 + 15| + C$$

Step 5: Final Answer: the correct option is C. $3 \ln |x^5 + 15| + C$

Question 3

Evaluate $\frac{d}{dx}(\sin x)^{\ln x}$.

- a. $(\ln x)(\sin x)^{(\ln x)-1} \cdot (\cos x)$
- b. $(\sin x)^{\ln x} \cdot \ln(\sin x) \cdot \frac{1}{x}$
- c. $(\sin x)^{\ln x} \left(\frac{1}{x} \cdot \ln(\sin x) + \ln x \cdot \cot x \right)$
- d. I do not know

Solution: c) $(\sin x)^{\ln x} \left(\frac{1}{x} \cdot \ln(\sin x) + \ln x \cdot \cot x \right)$

Step 1: Use logarithmic differentiation

Step 1.1:

Rewrite the expression as a variable $y = (\sin x)^{\ln x}$

Step 1.2:

Take natural log on both sides $\ln(y) = \ln((\sin x)^{\ln x})$

Step 1.3:

Apply the log power rule **Log Power Rule:** $\ln(a^b) = b \ln(a)$.

Step 1.4:

Find $\ln((\sin x)^{\ln x})$ $\ln((\sin x)^{\ln x}) = (\ln x) \ln(\sin x)$

Step 2: Differentiate both sides of the equation

Step 2.1:

Differentiate $\ln y$ using chain rule **Chain Rule:** if functions f and g are both differentiable and $F(x) = f(g(x))$, then $F'(x) = f'(g(x))g'(x)$

Step 2.2:

Apply the chain rule $\frac{d}{dx}(\ln y) = (\ln y)'(y)'$

Step 2.3:

Evaluate $(\ln y)'(y)'$ $(\ln y)' = \frac{1}{y}$ and $y' = \frac{dy}{dx}$, so $(\ln y)'(y)' = \frac{1}{y} \frac{dy}{dx}$

Step 2.4:

Differentiate $(\ln x) \ln(\sin x)$ using product rule **Product Rule:** if functions f and g are both differentiable and $F(x) = f(x)g(x)$, then $F'(x) = f(x)g'(x) + f'(x)g(x)$

Step 2.5:

Apply the product rule

$$\frac{d}{dx}(\ln x) \ln(\sin x) = (\ln x)' \ln(\sin x) + (\ln x) \ln(\sin x)'$$

Step 3: Evaluate $(\ln x)' \ln(\sin x) + (\ln x) \ln(\sin x)'$

Step 3.1:

Differentiate $\ln x$

$$(\ln x)' = \frac{1}{x}$$

Step 3.2:

Differentiate $\ln(\sin x)$ **using**
chain rule

$$(\ln(\sin x))' = (\ln'(\sin x))(\sin x)'$$

Step 3.3:

Differentiate $\sin x$

$$(\sin x)' = \cos x$$

Step 3.4:

Combine the results

$$(\ln(\sin x))' = \frac{\cos x}{\sin x} = \cot x$$

Step 3.5:

Substitute derivatives back
into the product rule

$$\frac{d}{dx}[(\ln x) \ln(\sin x)] = \frac{1}{x} \ln(\sin x) + (\ln x) \cot x$$

Step 4: Solve for $\frac{dy}{dx}$

Step 4.1:

Set both derivatives equal

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{x} \ln(\sin x) + (\ln x) \cot x$$

Step 4.2:

Multiply both sides by y

$$\frac{dy}{dx} = y \left(\frac{1}{x} \ln(\sin x) + (\ln x) \cot x \right)$$

Step 4.3:

Replace y **with** $(\sin x)^{\ln x}$

$$\frac{d}{dx}(\sin x)^{\ln x} = (\sin x)^{\ln x} \left(\frac{1}{x} \ln(\sin x) + (\ln x) \cot x \right)$$

Step 5: Final Answer: the correct option is C. $(\sin x)^{\ln x} \left(\frac{1}{x} \cdot \ln(\sin x) + \ln x \cdot \cot x \right)$

Question 4

Evaluate $\int x^3 \cdot 5^{x^4} dx$.

- a. $\frac{5^{x^4} \ln 5}{4} + C$
- b. $\frac{x^4 \cdot 5^{x^4}}{4} + C$
- c. $\frac{5^{x^4}}{4 \ln 5} + C$
- d. I do not know

Solution: c) $\frac{5^{x^4}}{4 \ln 5} + C$

Step 1: Use U-Substitution, Let $u = x^4$

Step 1.1:

Differentiate both sides of the equation $\frac{du}{dx} = \frac{d}{dx}(x^4)$

Step 1.2:

Apply the power rule **Power Rule:** if $g(x) = x^n$, then $g'(x) = nx^{n-1}$.

Step 1.3:

Differentiate x^4 $\frac{d}{dx}(x^4) = 4x^{4-1} = 4x^3$

Step 1.4:

Combine the results $\frac{du}{dx} = 4x^3$

Step 2: Convert the equation into differential form

Step 2.1:

Multiply both sides by dx $du = 4x^3 dx$

Step 2.2:

Solve for $x^3 dx$ Divide both sides by 4: $\frac{1}{4} du = x^3 dx$

Step 3: Rewrite the integral in terms of u

Step 3.1:

Substitute $u = x^4$ $5^{x^4} = 5^u$

Step 3.2:

Substitute $x^3 dx = \frac{1}{4} du$

$$\int x^3 \cdot 5^{x^4} dx = \int 5^u \cdot \frac{1}{4} du$$

Step 4: Evaluate $\int 5^u \cdot \frac{1}{4} du$

Step 4.1:

Use constant multiple rule

Constant Multiple Rule: $\int cf(x)dx = c \int f(x)dx.$

Step 4.2:

Apply the rule with $c = \frac{1}{4}$

$$\int 5^u \cdot \frac{1}{4} du = \frac{1}{4} \int 5^u du$$

Step 4.3:

Use exponential integral rule

Exponential Integral Rule: $\int a^u du = \frac{a^u}{\ln a} + C.$

Step 4.4:

Apply the rule with $a = 5$

$$\int 5^u du = \frac{5^u}{\ln 5} + C$$

Step 4.5:

Multiply by $\frac{1}{4}$

$$\frac{1}{4} \left(\frac{5^u}{\ln 5} + C \right) = \frac{5^u}{4 \ln 5} + C$$

Step 5: Substitute back to x

Step 5.1:

Replace u with x^4

$$\frac{5^u}{4 \ln 5} + C = \frac{5^{x^4}}{4 \ln 5} + C$$

Step 6: Final Answer: the correct option is C. $\frac{5^{x^4}}{4 \ln 5} + C$

Question 5

Evaluate $\int \frac{(\log_8 x)^2}{x} dx$.

- a. $(\ln 8) \cdot (\log_8 x)^2 + C$
- b. $\frac{\ln 8}{3} \cdot (\log_8 x)^3 + C$
- c. $\frac{(\ln 8)^2}{3} \cdot (\log_8 x)^3 + C$
- d. I do not know

Solution: b) $\frac{\ln 8}{3} \cdot (\log_8 x)^3 + C$

Step 1: Use U-Substitution, Let $u = \log_8 x$

Step 1.1:

Rewrite $\log_8 x$ using change of base formula

Change of Base Formula: $\log_b x = \frac{\ln x}{\ln b}$.

Step 1.2:

Apply the formula with $b = 8$ $u = \log_8 x = \frac{\ln x}{\ln 8}$

Step 2: Find du

Step 2.1:

Differentiate both sides of the equation $\frac{du}{dx} = \frac{d}{dx} \left(\frac{\ln x}{\ln 8} \right)$

Step 2.2:

Use constant multiple rule for derivatives

Constant Multiple Rule: $\frac{d}{dx}[cf(x)] = cf'(x)$.

Step 2.3:

Apply the formula with $c = \frac{1}{\ln 8}$ $\frac{d}{dx} \frac{\ln x}{\ln 8} = \frac{1}{\ln 8} \frac{d}{dx}(\ln x)$
 $c = \ln 8$

Step 2.4:

Differentiate $\ln x$ $\frac{d}{dx}(\ln x) = \frac{1}{x}$

Step 2.5:

Combine the results $\frac{du}{dx} = \frac{1}{\ln 8} \cdot \frac{1}{x} = \frac{1}{x \ln 8}$

Step 3: Convert into differential form

Step 3.1:

Multiply both sides by dx $du = \frac{1}{x \ln 8} dx$

Step 3.2:

Solve for $\frac{1}{x} dx$ Multiply both sides by $\ln 8$: $(\ln 8) du = \frac{1}{x} dx$

Step 4: Rewrite the integral in terms of u **Step 4.1:**

Substitute $u = \log_8 x$ $(\log_8 x)^2 = u^2$

Step 4.2:

Substitute $\frac{1}{x} dx = (\ln 8) du$ $\int \frac{(\log_8 x)^2}{x} dx = \int u^2 (\ln 8) du$

Step 4.3:

Use constant multiple rule **Constant Multiple Rule:** $\int cf(x) dx = c \int f(x) dx$.

Step 4.4:

Rewrite the integral $\int u^2 (\ln 8) du = (\ln 8) \int u^2 du$

Step 5: Evaluate $(\ln 8) \int u^2 du$ **Step 5.1:**

Apply power rule for integrals **Power Rule for Integrals:** if $n \neq -1$, $\int u^n du = \frac{u^{n+1}}{n+1} + C$.

Step 5.2:

Apply the rule with $n = 2$ $\int u^2 du = \frac{u^3}{3} + C$

Step 5.3:

Multiply by $\ln 8$ $(\ln 8) \left(\frac{u^3}{3} + C \right) = \frac{\ln 8}{3} u^3 + C$

Step 6: Substitute back to x **Step 6.1:**

Replace u with $\log_8 x$ $\frac{\ln 8}{3} u^3 + C = \frac{\ln 8}{3} (\log_8 x)^3 + C$

Step 7: Final Answer: the correct option is B. $\frac{\ln 8}{3} \cdot (\log_8 x)^3 + C$

Question 6

Evaluate $\lim_{x \rightarrow \infty} \frac{\ln(3x^2 + 7)}{\ln(5x^2 + 6)}$.

- a. $\ln\left(\frac{3}{5}\right)$
- b. 1
- c. $\frac{5}{3}$
- d. I do not know

Solution: b) 1

Step 1: Identify the limit form

Step 1.1:

Evaluate the numerator as $3x^2 + 7 \rightarrow \infty$, therefore $\ln(3x^2 + 7) \rightarrow \infty$
 $x \rightarrow \infty$

Step 1.2:

Evaluate the denominator as $5x^2 + 6 \rightarrow \infty$, therefore $\ln(5x^2 + 6) \rightarrow \infty$
 $x \rightarrow \infty$

Step 1.3:

Determine the indeterminate form The limit is of the form $\frac{\infty}{\infty}$
form

Step 2: Apply L'Hôpital's Rule

Step 2.1:

State L'Hôpital's Rule

L'Hôpital's Rule: If $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ is $\frac{0}{0}$ or $\frac{\infty}{\infty}$, then
 $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$

Step 2.2:

Differentiate numerator and denominator

$$\lim_{x \rightarrow \infty} \frac{\ln(3x^2 + 7)}{\ln(5x^2 + 6)} = \lim_{x \rightarrow \infty} \frac{\frac{d}{dx} \ln(3x^2 + 7)}{\frac{d}{dx} \ln(5x^2 + 6)}$$

Step 3: Differentiate the numerator

Step 3.1:

Use chain rule

Chain Rule: if $F(x) = \ln(h(x))$, then $F'(x) = \frac{h'(x)}{h(x)}$.

Step 3.2:

Differentiate inside function

$3x^2 + 7$

Power Rule: if $g(x) = x^n$, then $g'(x) = nx^{n-1}$.**Step 3.3:**

Apply the power rule

$$\frac{d}{dx}(3x^2) = 3 \cdot 2x^{2-1} = 6x$$

Step 3.4:

Differentiate the constant

$$\frac{d}{dx}(7) = 0$$

Step 3.5:

Combine the results

$$\frac{d}{dx}(3x^2 + 7) = 6x$$

Step 3.6:

Apply chain rule result

$$\frac{d}{dx} \ln(3x^2 + 7) = \frac{6x}{3x^2 + 7}$$

Step 4: Differentiate the denominator**Step 4.1:**

Apply chain rule again

$$\frac{d}{dx} \ln(5x^2 + 6) = \frac{(5x^2 + 6)'}{5x^2 + 6}$$

Step 4.2:Differentiate $5x^2$

$$5 \cdot 2x^{2-1} = 10x$$

Step 4.3:

Differentiate the constant

$$\frac{d}{dx}(6) = 0$$

Step 4.4:

Combine the results

$$\frac{d}{dx}(5x^2 + 6) = 10x$$

Step 4.5:

Apply chain rule result

$$\frac{d}{dx} \ln(5x^2 + 6) = \frac{10x}{5x^2 + 6}$$

Step 5: Rewrite the limit using derivatives**Step 5.1:**

Substitute derivatives into the limit

$$\lim_{x \rightarrow \infty} \frac{\frac{6x}{3x^2 + 7}}{\frac{10x}{5x^2 + 6}}$$

Step 5.2:

Rewrite complex fraction as multiplication

$$\lim_{x \rightarrow \infty} \left(\frac{6x}{3x^2 + 7} \cdot \frac{5x^2 + 6}{10x} \right)$$

Step 5.3:

Multiply numerators and
denominators

$$\lim_{x \rightarrow \infty} \frac{6x(5x^2 + 6)}{10x(3x^2 + 7)}$$

Step 5.4:

Distribute $6x$ in numerator

$$6x(5x^2 + 6) = 30x^3 + 36x$$

Step 5.5:

Distribute $10x$ in denominator

$$10x(3x^2 + 7) = 30x^3 + 70x$$

Step 5.6:

Rewrite the limit

$$\lim_{x \rightarrow \infty} \frac{30x^3 + 36x}{30x^3 + 70x}$$

Step 6: Evaluate the limit as $x \rightarrow \infty$

Step 6.1:

Factor highest power x^3 from
numerator and denominator

$$\frac{x^3(30 + \frac{36}{x^2})}{x^3(30 + \frac{70}{x^2})}$$

Step 6.2:

Cancel x^3

$$\frac{30 + \frac{36}{x^2}}{30 + \frac{70}{x^2}}$$

Step 6.3:

Evaluate as $x \rightarrow \infty$

$$\frac{36}{x^2} \rightarrow 0 \text{ and } \frac{70}{x^2} \rightarrow 0$$

Step 6.4:

Compute final value

$$\frac{30 + 0}{30 + 0} = 1$$

Step 7: Final Answer: the correct option is B. 1

Question 7

Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^3 + 5x^2}$.

- a. 0
- b. $\frac{1}{10}$
- c. $\frac{2}{5}$
- d. I don't know

Solution: c) $\frac{2}{5}$

Step 1: Identify the limit form

Step 1.1:

Evaluate numerator at $x = 0$

$$\begin{aligned} 1 - \cos 2x &= 1 - \cos(2 \cdot 0) \\ &= 1 - \cos(0) \\ &= 1 - 1 \\ &= 0 \end{aligned}$$

Step 1.2:

Evaluate denominator at $x = 0$

$$\begin{aligned} x^3 + 5x^2 &= (0)^3 + 5(0)^2 \\ &= 0 + 0 \\ &= 0 \end{aligned}$$

Step 1.3:

Determine indeterminate form $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^3 + 5x^2} = \frac{0}{0}$

Step 2: Apply L'Hôpital's Rule (First Time)

Step 2.1:

State L'Hôpital's Rule

L'Hôpital's Rule: If $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ gives $\frac{0}{0}$ or $\frac{\infty}{\infty}$, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}, \text{ provided the new limit exists.}$$

Step 2.2:

Differentiate numerator

$$\frac{d}{dx}(1 - \cos 2x) = \frac{d}{dx}(1) + \frac{d}{dx}(-\cos 2x)$$

Step 2.3:

Differentiate 1

$$\frac{d}{dx}(1) = 0$$

Step 2.4:

Differentiate $-\cos 2x$ using
chain rule

Chain Rule: If $F(x) = \cos(h(x))$, then $\frac{d}{dx}[\cos(h(x))] = -\sin(h(x))h'(x)$.

Step 2.5:

Differentiate inside function

$$\frac{d}{dx}(2x) = 2$$

Step 2.6:

Combine chain rule result

$$\frac{d}{dx}(-\cos 2x) = (-1)(-\sin 2x)(2) = 2 \sin 2x$$

Step 2.7:

Combine numerator
derivatives

$$\frac{d}{dx}(1 - \cos 2x) = 0 + 2 \sin 2x = 2 \sin 2x$$

Step 2.8:

Differentiate denominator

$$\frac{d}{dx}(x^3 + 5x^2) = \frac{d}{dx}(x^3) + \frac{d}{dx}(5x^2)$$

Step 2.9:

Apply power rule

Power Rule: If $g(x) = x^n$, then $g'(x) = nx^{n-1}$.

Step 2.10:

Differentiate x^3

$$\frac{d}{dx}(x^3) = 3x^2$$

Step 2.11:

Differentiate $5x^2$

$$\frac{d}{dx}(5x^2) = 10x$$

Step 2.12:

Combine denominator
derivatives

$$\frac{d}{dx}(x^3 + 5x^2) = 3x^2 + 10x$$

Step 2.13:

Rewrite limit

$$\lim_{x \rightarrow 0} \frac{2 \sin 2x}{3x^2 + 10x}$$

Step 3: Check the new limit form

Step 3.1:**Evaluate numerator at $x = 0$**

$$\begin{aligned}
 2 \sin 2x &= 2 \sin(2 \cdot 0) \\
 &= 2 \sin(0) \\
 &= 2(0) \\
 &= 0
 \end{aligned}$$

Step 3.2:**Evaluate denominator at $x = 0$**

$$\begin{aligned}
 3x^2 + 10x &= 3(0)^2 + 10(0) \\
 &= 0 + 0 \\
 &= 0
 \end{aligned}$$

Step 3.3:**Determine indeterminate form** $\lim_{x \rightarrow 0} \frac{2 \sin 2x}{3x^2 + 10x} = \frac{0}{0}$ **Step 4: Apply L'Hôpital's Rule (Second Time)****Step 4.1:****Differentiate numerator** $\frac{d}{dx}(2 \sin 2x) = 2(\cos 2x)(2) = 4 \cos 2x$ **Step 4.2:****Differentiate denominator** $\frac{d}{dx}(3x^2 + 10x) = 6x + 10$ **Step 4.3:****Rewrite limit** $\lim_{x \rightarrow 0} \frac{4 \cos 2x}{6x + 10}$ **Step 5: Evaluate the limit****Step 5.1:****Evaluate numerator at $x = 0$** $4 \cos 2x = 4 \cos(2 \cdot 0) = 4 \cos(0) = 4(1) = 4$ **Step 5.2:****Evaluate denominator at $x = 0$** $6x + 10 = 6(0) + 10 = 10$ **Step 5.3:****Combine the result** $\lim_{x \rightarrow 0} \frac{4 \cos 2x}{6x + 10} = \frac{4}{10} = \frac{2}{5}$ **Step 6: Final Answer: the correct option is C.** $\frac{2}{5}$

Question 8

Evaluate $\lim_{x \rightarrow \infty} \frac{x^{25}}{2^x}$.

- a. 0
- b. $\frac{25!}{(\ln 2)^{25}}$
- c. ∞
- d. I don't know

Solution: a) 0

Step 1: Identify the limit form

Step 1.1:

Evaluate numerator as $x \rightarrow \infty$ $x^{25} \rightarrow \infty$

Step 1.2:

Evaluate denominator as $2^x \rightarrow \infty$
 $x \rightarrow \infty$

Step 1.3:

Determine indeterminate form $\lim_{x \rightarrow \infty} \frac{x^{25}}{2^x} = \frac{\infty}{\infty}$

Step 2: Apply L'Hôpital's Rule (First Time)

Step 2.1:

L'Hôpital's Rule

L'Hôpital's Rule: If $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ gives $\frac{0}{0}$ or $\frac{\infty}{\infty}$, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}.$$

Step 2.2:

Differentiate numerator

Power Rule: If $g(x) = x^n$, then $g'(x) = nx^{n-1}$.

Step 2.3:

Apply power rule

$$\frac{d}{dx}(x^{25}) = 25x^{24}$$

Step 2.4:

Differentiate denominator

Exponential Derivative Rule: $\frac{d}{dx}(a^x) = a^x \ln a$.

Step 2.5:

Apply exponential rule

$$\frac{d}{dx}(2^x) = 2^x \ln 2$$

Step 2.6:

Rewrite limit

$$\lim_{x \rightarrow \infty} \frac{25x^{24}}{2^x \ln 2}$$

Step 3: Apply L'Hôpital's Rule (Second Time)

Step 3.1:

Differentiate numerator again $\frac{d}{dx}(25x^{24}) = 25 \cdot 24x^{23}$

Step 3.2:

Differentiate denominator again $\frac{d}{dx}(2^x \ln 2) = \ln 2 \cdot (2^x \ln 2) = 2^x (\ln 2)^2$

Step 3.3:

Rewrite limit

$$\lim_{x \rightarrow \infty} \frac{25 \cdot 24x^{23}}{2^x (\ln 2)^2}$$

Step 3.4:

Continue applying L'Hôpital After repeating this process 25 times

Step 3.5:

Final differentiated form

$$\lim_{x \rightarrow \infty} \frac{25!}{2^x (\ln 2)^{25}}$$

Step 4: Evaluate the limit

Step 4.1:

Analyze numerator $25!$ is a constant.

Step 4.2:

Analyze denominator $2^x (\ln 2)^{25} \rightarrow \infty$

Step 4.3:

Evaluate limit

$$\lim_{x \rightarrow \infty} \frac{25!}{2^x (\ln 2)^{25}} = \frac{\text{constant}}{\infty} = 0$$

Step 5: Final Answer: the correct option is A. 0